STRATEGIC BRIEFING

Autonomous AI Collections

Scaling Compliant, Fair, and Empathetic Debt Interventions using Agentic AI at Geldium Finance

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System Architecture

Based on the exploratory data analysis (EDA) and predictive models built in our earlier diagnostic phase, this architecture puts customer insights into production-level automation.

1. Ingest

Ingests customer records, applying automatic utilization boundaries and median-imputed income pools.

2. Predict

XGBoost maps stress factors and engineered late-payment counts to forecast default probability.

3. Automate

AI agent triggers early notifications or transfers complex accounts to specialized hardship streams.

4. Learn

Outcome tracking monitors payment success to continually retrain and optimize risk thresholds.

Operational Decision Logic

A dual-engine framework grounded in our earlier data quality analysis to address known anomalies while maximizing predictive yield.

Heuristic Business Rules

Hard boundaries configured to intercept data quality errors and protect customer interactions.

- **Utilization Boundaries:** Instantly caps credit utilization values to 100% to resolve anomalous over-ceiling logs.
- **Tenure Intercepts:** Automatically flags zero-tenure records with active history for systemic reconciliation.
- **Manual Dispute Holds:** Freezes automated agent outreach immediately upon formal complaint lodging.

Machine Learning Probabilities

Model predictions designed to capture complex, non-linear risk interactions across verified financial fields.

- **Synergistic Risk Capture:** Identifies high-risk indicators when utilization exceeds 85% alongside a high DTI.
- **Temporal Behavior Tracking:** Synthesizes cumulative categorical variables from the 6-month payment windows.
- **Proactive Reductions:** Optimizes operational risk by trigger-lowering credit lines down to actual balances.

Role of Agentic AI

Operational Layer	Autonomous Agent Execution	Human-in-the-Loop Oversight
Outreach & Communications	Generates and sends automated SMS alerts, emails, and WhatsApp payment reminders.	Reviews and approves core copy templates, tone guidelines, and multi-channel strategies.
Hardship Setup	Validates basic self-service hardship plans and processes simple interest deferrals.	Reviews custom structured restructuring cases, bankruptcy declarations, and asset liquidations.
Dispute Resolutions	Catalogs inbound dispute claims and dynamically freezes outbound messaging.	Directly investigates customer disputes, logs claims errors, and manually updates profiles.

Ethical & Responsible Guardrails

1. Active Fairness

Enforces a **Disparate Impact Ratio ≥ 0.80** across locations and age brackets to prevent biased credit outcomes.

2. SHAP Explainability

Translates complex ensemble calculations into individual feature impact weights, making risk scores transparent.

3. Compliant Auditing

Aligned with ECOA, FCRA, and GDPR parameters. Preserves full, non-technical auditable trails for regulatory exams.

Explaining Decisions Simply

SHAP

POST-HOC EXPLAINABILITY

Demystifying our XGBoost Model's Risk Scores

Instead of presenting complex mathematical tree paths, our framework leverages SHAP weights to isolate the exact drivers of predictive risk for non-technical teams.

Top Predictive Feature Importances:

- Unfavorable_Payment_Count: **+0.38**
- Credit_Utilization: **+0.29**
- Debt_to_Income_Ratio: **+0.21**
- Account_Tenure: **-0.14**

Non-Technical Customer Support Script: *"This customer's risk profile increased primarily due to missing multiple payments and over-utilizing their limit, though their stable account tenure slightly helped balance their risk."*

Expected Performance Impact

Visualizing delinquency concentrations predicted by our trained risk model across actual customer segments.

These risk profiles are derived directly from our XGBoost modeling results, optimized to achieve an operational baseline ROC-AUC of 0.82 and F1-Score balance under imbalanced classes.

Strategic Outcome Summary

Quantitative Financial KPIs

Key financial improvements derived directly from our SMART business justification targets.

- **Roll-Rate Mitigation:** Target a 25% decrease in transition rates from early-stage (30 DPD) to serious delinquency (90+ DPD).
- **Operational Optimization:** Relieve 40% of manual team constraints through automated agent messaging flows.
- **Portfolio Protection:** Mitigates write-offs by preemptively triggering account limitations based on our 90-day pipeline roadmap.

Qualitative Customer Trust

Strengthening client relationships through fair, personalized, and proactive support pathways.

- **Proactive Hardship Offers:** Integrates automated self-service portals to support under-pressure cardholders.
- **Equitable Underwriting:** Standardizes bias assessments using demographic disparity tracking.
- **Actionable Explanations:** Employs clear decision disclosures to maintain customer trust during limit modifications.